

AI Research Roadmap

GOAL

Become an AI Researcher — able to read, implement, and publish research.

THE PATH

ML Engineer → **Research Engineer** → **AI Researcher**

PHASE 1

Engineer Foundations

≈ 9–12 months

Math → Python data stack →
Machine Learning → Deep Learning
→ Projects. Ends: job-ready ML
Engineer.

PHASE 2

Research Depth

≈ 6–9 months

Rigorous math → Transformers &
LLMs → reading and reproducing
research papers.

PHASE 3

Research Practice

Ongoing

Original experiments → writing &
publishing → research community
contributions.

More hours = faster.

Four rules, one map

Simple system: numbered steps, one resource each, done strictly in order.

THE FOUR GOLDEN RULES

1. **Strict order, no parallel learning.** Start a step, finish it completely, then start the next.
2. **One concept = one resource.** Never two courses for the same topic.
3. A video counts as done only when you've **passed the mentor test** on it. Testing is the course.
4. **Optional:** publish what you build to GitHub + your portfolio — recommended when job hunting, but your choice.

Phase 1 · Engineer Foundations

Five stages, nine steps, in order. Full details with links on pages 3–4.

Stage	What you learn	Where
1 · Math	Linear Algebra, Calculus, Multivariable basics, Probability & Statistics	3Blue1Brown + StatQuest (free)
2 · Data Analysis	Python data stack: NumPy, Pandas, Matplotlib, EDA	freeCodeCamp (free)
3 · Machine Learning	Supervised, unsupervised & reinforcement learning	Coursera — Andrew Ng
4 · Deep Learning	Neural networks, CNNs, sequence models + GPT from scratch	Coursera + Karpathy
5 · Projects	Three portfolio builds, beginner → advanced	Your GitHub

1 Math short videos + mentor tests · steps 1–4, one at a time

Method: watch the short video → tell your mentor "done" → take a **3-question test** (easy · normal · hard) → solve on paper → get your score, the error named, and a level. **No long lectures.**

#	Concept	Resource (one only)
1	Linear Algebra Vectors · span · independence · basis · matrix transformations · inverses · eigenvalues Chapters: 16 — Must do: 1–9 and 13–15. Skip 10–12 (cross products, Cramer's rule)	3Blue1Brown — Essence of Linear Algebra Open →
2	Calculus Limits · derivatives · power/product/quotient rules · chain rule (= backpropagation) Chapters: 12 — Must do: 1–7 (through limits & the chain rule). Chapters 8–12: skip	3Blue1Brown — Essence of Calculus Open →
3	Multivariable basics Partial derivatives · the gradient (what gradient descent uses) Topics: 2 only — taught directly in the mentor chat with worked examples and tests	Mentor-taught + 3Blue1Brown gradient videos Open →
4	Probability & Statistics Probability · conditional probability & Bayes · random variables · distributions · mean & variance Short videos, 5–15 min each — Must do: descriptive stats, probability, Bayes' theorem, random variables, distributions	StatQuest — Statistics Fundamentals Open →

2 Data Analysis you already have Python basics — go straight to the data stack

#	Concept	Resource (one only)
5	Python data stack Jupyter · NumPy · Pandas · Matplotlib · EDA — includes 5 build projects (first portfolio commits)	freeCodeCamp — Data Analysis with Python Open →

3 Machine Learning the classic core

#	Concept	Resource (one only)
6	Machine Learning Regression · classification · decision trees · clustering · recommenders · reinforcement learning · best practices	Coursera — Machine Learning Specialization (Andrew Ng) · audit free Open →

4 Deep Learning the modern engine

#	Concept	Resource (one only)
7	Deep Learning Deep networks · optimization · train/test & variance analysis · CNNs · sequence models	Coursera — Deep Learning Specialization (Andrew Ng) · audit free Open →
8	★ Build GPT from scratch Backpropagation → your own mini-GPT, all in code — the key step toward your goal	Neural Networks: Zero to Hero (Andrej Karpathy) · free Open →

Framework rule: the courses use TensorFlow — fine. Your personal framework is **PyTorch**, the language of research (pytorch.org/tutorials).

5 Projects step 9 — build in order, ship each one

Level	Project	What it proves
Beginner	House Price Prediction	Regression, data cleaning, features
Intermediate	Movie Recommendation System	Recommenders, filtering
Advanced	Mini-GPT — train your own small language model, then fine-tune an open model	Transformers, training, evaluation

Finishing this page = job-ready ML Engineer. Publishing each project to GitHub + a short write-up: optional, but recommended.

From engineer to research-capable

Start only after Phase 1 is complete. Same system: one step, one resource, in order.

#	Concept	Resource (one only)
1	Linear Algebra — rigor pass Now you're ready for Strang. Problem sets with solutions — the psets are the point	MIT 18.06SC — Gilbert Strang (self-study version) Open →
2	Math of ML papers Vector calculus · probability · optimization — the notation real papers use	Mathematics for Machine Learning — free book Open →
3	NLP & Transformers Word vectors → attention → transformers → pretraining & LLMs	Stanford CS224n — free lectures on YouTube Open →
4	Hands-on LLMs Tokenizers · datasets · fine-tuning · evaluation · sharing models	Hugging Face — LLM Course · free Open →
5	★ Reproduce 3–5 papers Read → reimplement in PyTorch → match a reported number → write up what didn't match, on your blog	Papers with Code Open →

First five papers, read in this order: Attention Is All You Need → GPT-2 → BERT → Adam → LoRA.

Finish these five steps → you now operate like a Research Engineer

Doing research

Ongoing. Original work, in public, with a community.

The research loop

Reproduce → **ablate** (change one thing, measure the effect) → **question** → **small original experiment** → **write it up** → **share** → **repeat**. Original research is this loop run honestly, many times.

★ Your unfair advantage — use it

Native **Somali** speaker + health-domain expertise + real low-resource data context. Somali is genuinely underserved in NLP, and low-resource-language research is an active frontier. **Masakhane** publishes real, cited research with contributors exactly like you — no PhD gate. This is the most realistic route to a first publication.

[Open Masakhane →](#)

[EleutherAI →](#)

Also counts as research signal: open-source contributions (Hugging Face ecosystem, PyTorch), public paper notes, and benchmarks or evaluations published on your blog.

Milestone checklist

- ☐ Essence of Linear Algebra finished — all mentor tests passed **P1**
- ☐ freeCodeCamp Data Analysis certification earned **P1**
- ☐ ML + Deep Learning Specializations completed **P1**
- ☐ Zero to Hero done — your own mini-GPT trained **P1**
- ☐ 3 projects completed — publishing to GitHub optional **P1**
- ☐ 18.06SC problem sets completed **P2**
- ☐ First paper reproduction published on your blog **P2**
- ☐ First original experiment written up and shared **P3**
- ☐ First community contribution — Masakhane or open source **P3**

BOTTOM LINE

Finish **Phase 1** → job-ready ML Engineer.

Finish **Phase 2** → you operate like a Research Engineer.

Phase 3 is research itself. One step at a time, one resource per step, strictly in order.